

ZAYAAN BHANWADIA

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EDUCATION

University of Toronto

Sept. 2025 – June 2028

HBSc in Computer Science – Software Engineering Specialist, Statistics Minor (Co-op) | GPA: 3.7/4.0

- Activities: 3x Hackathon Winner, Web Developer @ UTRA, VP Technology @ CS Student Association
- Relevant Coursework: Software Design, Systems Programming, Data Structures & Algorithms, Computational Theory

EXPERIENCE

Software Developer Intern (*Full Stack*)

Sept. 2026 – Dec. 2026

FreshBooks

Toronto, ON

- Migrated **10+** legacy Flask endpoints to centralized FastAPI services, modernizing core banking infrastructure.
- Refactored banking locks, enabling concurrent updates across **5+** financial record types without cross-entity blocking.
- Extended SQLAlchemy models and APIs with consent-expiry tracking across **2** bank connection providers.
- Unified Plaid and Yodlee flows across backend services and Ember.js, supporting staged rollout to **5,000+** users.

Autonomy Software Engineer (*Mapping & Planning*)

July 2026 – Present

University of Toronto Formula Racing – Driverless

Toronto, ON

- Developed mapping and path-planning software in C++ and ROS 2 for **Canada's first driverless FSAE car**.
- Built a SLAM pipeline mapping new racetracks to within **18 cm** accuracy using Kalman filters and factor graphs.
- Rebuilt the path planner to keep **96%** of paths on track in simulation, fixing sharp turns that stalled the old system.

Software Engineer Intern (*Backend AI*)

July 2026 – Sept. 2026

FlyRank AI

Toronto, ON

- Built a centralized embeddable widget platform, routing form submissions to a dashboard via a **one-line** script tag.
- Secured a public endpoint with validation, rate limiting, and geolocation checks, cutting spam by **80%**.
- Prevented duplicate charges under **50+** simultaneous requests by deduplicating payment events in the database.

Software Developer Intern (*Full Stack*)

May 2026 – July 2026

KorraNet Creative

Remote

- Optimized Meta OAuth token handling and storage, reducing token refresh failures by **40%+** for AI integrations.
- Reduced system crashes and failed API calls by **85%+** by adding API error handling and timeout prevention.
- Migrated hardcoded secrets to Firebase Secret Manager, enabling key rotation without redeploying services.

PROJECTS

Embedded Key-Value Store | *C++, CMake, Multithreading, Linux*

- Built an embedded database with **16-way** sharded locking, boosting read throughput **5x (8M to 40M ops/sec)**.
- Implemented reader-writer locks so reads proceed during disk writes, preventing race conditions across **16 threads**.
- Created a file compaction routine to safely clean up keys, reducing disk usage by **45%+** without pausing the app.
- Guaranteed crash recovery by persisting every update to a checksummed write-ahead log before applying changes.

Context Sync Extension | *TypeScript, VS Code Extension API*

- Developed a VS Code extension with **700+ downloads** that automates AI chat context sharing across environments.
- Reduced token use by **25%+** by modelling chats as a weighted graph and selecting context via shortest-path search.
- Compacted chat summaries by **30%+** by designing a Markdown schema to pack more context into fewer tokens.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C++, C, TypeScript, SQL, Bash, Ruby, HTML/CSS

Libraries/Frameworks: FastAPI, Express.js, Node.js, React.js, Ember.js, SQLAlchemy, ROS 2, YOLO

Developer Tools: Linux, Docker, Git, GitHub, Claude Code, RabbitMQ, Jira, Confluence

Cloud/Databases: PostgreSQL, MySQL, AWS, Azure, GCP, Vercel, Vultr, Supabase